

Conceptualising Interaction, Cognitive Aspects and Social Interaction

1.

Transitioning to the Design Space

(19) The move from the problem space, characterised by agricultural waste and food insecurity, to the design space requires identifying a specific focus area that addresses the “pent-up anger” and frustration previously identified in the Eastern Cape. By conceptualizing interaction, we define a framework of general concepts that outline what stakeholders can do with the proposed radio system from Assignment 01, and what they need to understand to use it effectively.

↳ A purely “radio system” is not an ideal digital solution.

The Rural Context of Use

The potential solution operates within the specific environmental and social realities of the Eastern Cape. Subsistence farmers will interact with the radio system while working in fields or resting at home, using simple mobile phones to share information via SMS or voice calls(directed to the station).

Transporters will engage with the system while mobile, listening to the radio in their vehicles to identify delivery routes. Because these users are often engaged in physically demanding tasks, an audio-based interface is far more appropriate than a high-attention digital application

Other users?
-2

Usability Goals and User Experience (UX)

The proposed system aims to fundamentally transform the current “frustrating” and “inefficient” user experience into one that is “satisfying” and “empowering”

} Discuss - Which UX for which user? Why & how?
-3

- **Effectiveness and Utility:** The system must bridge the information gap, ensuring that surplus produce actually reaches households in need, thereby fulfilling its **primary purpose**.

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Assignment 02 Draft

Due Date: 03 August 2026

- **Efficiency:** By establishing a fixed and predictable broadcast slot, the design eliminates the need for stakeholders to rely on unreliable verbal communication or fruitless individual searches for buyers
- **Safety and Trust:** By leveraging trusted intermediaries such as local radio hosts and spaza shop owners, the system provides a credible environment for trade.

Tasks Users need to Complete

The conceptual model supports several critical tasks that users must complete to ensure the flow of produce:

- **Data Submission (Farmer):** Listing surplus crop types, quantities, and locations via a standard mobile communication *such as?* - 2
- **Information Synthesis (Radio host):** organising incoming data from farmers into a coherent broadcast. *There is no "digital system" where the radio host is a "user"*
- **Procurement (Trader/household):** Identifying available local stock and obtaining contact details from the broadcast ✓
- **Logistics coordination (Transporter):** Responding to delivery requests to move produce from fields to markets/households.

Leveraging Pre-existing Knowledge

In line with the principle of "Designing for implementers", the system requires minimal new technical knowledge. It relies on existing technological literacy (Knowing how to use a basic phone and radio) and social literacy (familiarity with local languages and community networks). By integrating the system into established daily routines, the design minimizes cognitive load and ensures high learnability. ✓

Interface metaphors

Marketplace, the radio broadcast slot also functions like a community marketplace where the farmer can just put up a stall to sell, by announcing their surplus produce and listeners are like people who pass by the stall, listeners like spaza owners who respond are like people who buy at the stall. The radio is the marketplace space, the farmers are sellers, spaza owners are buyers.

Noticeboard, the radio broadcast is like a public noticeboard where farmers write their announcements regarding the availability of the produce. The spaza owners and the

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transport providers read the noticeboard by always tuning in, at a fixed time of the farmer's hour. ✓

Interaction types

Conversing, farmers sometimes call in live or get invited during the broadcast slot where they announce their produce quantity. Spaza owners and transport owners can also call in to respond which creates a conversational interaction.

So no interaction with a system.

-4

Instructing, The farmer instructs the system which is the radio broadcast by calling in or sending an SMS with all the surplus details then the radio presenters broadcast these instructions to the wider community.

so the "system" does not follow the instruction? Then not instructing.
This is not interaction with the "proposed solution".

2. a) The name of the AI tool that i have used is Gemini ai

The prompt i have submitted

Provide a solution that can help connect farmers with food surplus produce in the rural areas, with households facing hunger and in need of affordable food, transport providers, spaza shop owners in a different area, describe the tasks that the system support and explain how the solution addresses the usability goals (the usability goals include Effectiveness, efficiency, safety, utility, learnability, memorability and satisfaction).

The full GenAI response

Proposed Solution: AgriBridge Ecosystem

AgriBridge is a lightweight, hybrid digital and USSD platform that connects rural smallholder farmers with urban/semi-urban spaza shop owners, local transport providers, and food-insecure households.

Because rural areas often face poor connectivity and lower smartphone adoption, the system works via a dual interface: a **mobile smartphone app** (for spaza owners and transport operators) and an **unstructured supplementary service data (USSD) / WhatsApp menu** (for rural farmers and low-income households).

Core Tasks Supported by the System

- 1. **Surplus Produce Listing (Farmers)**
 - Farmers list surplus produce by entering item type, estimated quantity, harvested date, and price via USSD (*120*#) or WhatsApp audio notes.
- 2. **Aggregated Inventory & Micro-Wholesale (Spaza Shop Owners)**
 - Spaza shop owners view nearby surplus produce listings, pool orders with other local vendors to achieve bulk pricing, and purchase stock at lower rates than commercial wholesalers.
- 3. **Dynamic Route Optimization & Load Matching (Transport Providers)**
 - Transport operators (e.g., local bakkie owners, taxi drivers on return trips) receive job alerts for picking up produce from farms and delivering it to spaza shops or community hubs. They accept bids and get optimized drop-off routes.
- 4. **Discounted Food Vouchers & Community Hub Pickups (Households)**
 - Low-income households receive SMS digital vouchers (via NGO/government subsidies or local community initiatives) to redeem for affordable fresh produce directly at participating spaza shops or community collection points.
- 5. **Quality & Temperature Verification (System Automated)**
 - Transporters and spaza owners scan a QR code/PIN code upon pickup and drop-off to log produce condition, maintaining safety and traceability along the chain.

How the Solution Addresses Usability Goals

Usability Goal	System Strategy & Implementation
Effectiveness	Goal-Driven Workflows: Farmers successfully list crops in under 2 minutes using voice prompts in local languages or simple numerical USSD menus. Spaza owners accurately receive matched produce orders, ensuring zero lost orders and reducing rural food waste.
Efficiency	Automated Load Pooling: The system automatically clusters orders from multiple spaza shops in the same geographic area to fill a single transport vehicle, minimizing travel time and reducing transport costs per unit.

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Safety	Error Prevention & Verification: High-friction financial transactions are secured using two-factor PIN codes upon delivery. Perishable freshness checks are enforced through required photo/voice confirmations before payment release, preventing food spoilage disputes.
Utility	Multi-Stakeholder Value: Provides farmers with a direct revenue stream, transport providers with steady delivery jobs, spaza shops with higher profit margins on fresh produce, and households with affordable nutrition nearby.
Learnability	Intuitive Interfaces: Uses recognized icon designs, localized audio instructions, and standardized WhatsApp chat flows so non-tech-savvy farmers and household users require zero formal training to start.
Memorability	Consistent UI & Shortcuts: Uses simple visual patterns and memorable shortcodes (e.g., repeating navigation options across USSD menus like 0 = Back, 9 = Home), making it effortless to re-use even after long gaps between harvests.
Satisfaction	Transparent & Fair Compensation: Immediate mobile money payout trigger upon verified delivery builds trust across all participants. Simple feedback/rating badges give transport providers and farmers pride and visible reputations.



b) Critique

Most of the tasks provided by GenAI are relevant but there are additional tasks that i think are also relevant, the first task is Damage management and control, the produce can

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spoil on the way while drivers are transporting the products, so the spaza owners must be able to report damaged products which can be through voice notes and the system must allow pictures for evidence of the damaged produce. Another task that is relevant is grading of freshness and quality where the farmer can indicate either using photos or inform the spaza owners through the SMS the quality of stock the farmer has (is it Grade A or a processing grade) so that the spaza owners know what to expect and where are they are spending money on. This prevents the downstream of rejections by spaza owners and the products do not have to go back and forth. The last task that is also relevant is Household voucher allocation control, the system did not include how the vouchers are loaded, distributed or verified as to which households deserve to get the voucher and how they redeem the voucher. ✓

Usability goals

All the Usability goals identified by the GenAI maps on to the ones that we have studied From **Effectiveness** which is about how good the system is in what it does, farmers send USSD or voice notes which reach spaza owners and they receive matched produce with no orders lost.

Efficiency is the way the system supports the user to carry out tasks, the system automatically collects orders from multiple shops in the same area and fills a single transport vehicle which helps them to reduce costs and travelling time.

Safety protects the user from dangerous conditions, the transaction is secured by a pin which is only known by the spaza owner upon delivery and it also requires a photo/video as confirmation for payments.

Utility is an extent to which the system provides the right functionality to users so they can do what they want, the farmers get market access, drivers get steady delivery jobs, spaza owners get produce at cheaper prices and households access food affordably.

Learnability is about how easy it is to learn to use the system, the system uses designs that they are familiar with so they don't need any training for it.

Memorability, how easy it is to remember how to use the system once you have learned how to use it, the system has shortcuts and consistent UI like 0= back which make it easy to re-use it.

Satisfaction, how enjoyable it is to use the system, it gives simple feedback and ratings to transport drivers and farmers which boosts their reputations. ✓

3.

Cognitive processes that are most relevant to our proposed solution

Attention Although the identified processes may be relevant, you do not have any digital product with a UI that enables users to interact with the product /solution.

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Attention is the cognitive process that enables users to focus on important information while filtering out irrelevant information. It involves choosing the things to focus on at a particular moment, allowing you to pay attention to information that is relevant to what you are doing. In our proposed radio broadcasting system, each stakeholder must pay attention for different reasons.

There is no UI for the farmer to interact with. The same holds for
Farmers need to listen carefully to confirm that their surplus produce has been announced *traders, transport providers etc.* correctly and that details such as the type of produce, quantity, location and contact information are accurate. Traders must pay attention so they can identify available produce that meets their business needs, note the farmer's contact details and arrange purchases before the produce is claimed by others. Transport providers need to listen for announcements that require transport services so they can determine where produce must be collected and delivered. Vulnerable households should pay attention to learn which farmers have surplus food available, where it can be accessed and how to contact the farmer if the produce is available for donation or sale at an affordable price.

Since radio broadcasts are temporary and information cannot be viewed again immediately, announcements should be short, clear and structured. Radio presenters should speak at a moderate pace, repeat important details such as contact numbers and locations and use a consistent announcement format for every broadcast. *Not UI design concerns.* These design considerations *not UI* help users capture the necessary information, reduce misunderstanding and improve the overall user experience.

Memory

Memory is the user's ability to retain and recall information when performing tasks. The success of our proposed solution depends on users remembering the daily broadcast schedule, the radio station and the important details announced during the programme. *Not related to UI design. Where is the interaction being designed?* The broadcast should be scheduled at the same time each day, to help memory by becoming part of a predictable routine. Farmers should receive confirmation messages after their announcements have aired. Moreover, important information like contact numbers and locations should be repeated during the broadcast. Consistent scheduling and repeated information make it easier for users to remember when to tune in and how to respond, creating a more reliable and usable system.

Interdependencies between Attention and Memory

Attention and memory are closely linked because users must first pay attention to the radio broadcast before they can remember the information that is shared. If listeners miss part of an announcement because they were distracted or the information was presented

too quickly, they may not remember the farmer's contact details or the available produce. Likewise, when the programme is broadcast consistently at the same time each day, users begin to remember the schedule, making it easier to pay attention when the broadcast starts. By designing broadcasts that encourage attention and support memory through repetition and consistency, the radio system becomes easier to use, more reliable and provides a more satisfying user experience for all stakeholders.

The "users" do not interact with the proposed solution.

4.

a) The multiplayer game we chose is Among Us. The game is played by 4-15 players. Most players are Crewmates, while one or more players are Impostors. The Crewmates' goal is to identify and vote the impostors out. On the other hand, Impostors aim to secretly eliminate Crewmates while avoiding being suspected until they outnumber the remaining Crewmates.

(5)

b) Social protocols and conventions applied in the game

Among us has a number of social protocols and conventions that encourage fair play and team work. During emergency meetings, players give each other a chance to speak, sharing observations and discussing evidence before voting. Crewmates are expected to collaborate by completing tasks and identifying the impostor. Another important convention is that eliminated players should not reveal the identity of the impostor so that the game remains fair for those who are still playing.

Number correctly!

c) Interaction design

How players interact and communicate

to type text? shortcuts?

Players interact with the game using keyboard and mouse to move around, complete tasks, report dead bodies and call emergency meetings. Communication takes place mostly during meetings via text chat or external voice chat platforms and communication is restricted outside these meetings.

Such as?
So no comms other than in meetings?

-3

Social phenomena that could occur in the game

A social phenomena that could occur in the game is Online deception. Because players cannot see each other's body language, facial expressions or emotional reactions, it may be simpler for them to lie and trick others. Nonverbal indicators like eye contact, tone of voice and gestures are frequently used in traditional face-to-face conversations to assess someone's sincerity. In Among Us, users often hide behind avatars and aliases and communicate via text chat or limited voice contact. Because most participants don't know each other personally, it's even easier for impostors to influence people and tell

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compelling lies without getting caught. This phenomenon happens because social cues are diminished and a degree of anonymity that is absent in in-person settings is created in the online environment. ✓